

Redacted



PATIENT HISTORY:

CHIEF COMPLAINT/PRE-OP/POST-OP DIAGNOSIS: Thyroid nodule
PROCEDURE: Not answered
SPECIFIC CLINICAL QUESTION: Not answered
OUTSIDE TISSUE DIAGNOSIS: Not answered
PRIOR MALIGNANCY: Not answered
CHEMO RADIATION: Not answered
ORGAN TRANSPLANT: Not answered
IMMUNOSUPPRESSION: Not answered
OTHER DISEASES: Not answered

ICD-0-3
Carcinoma, encapsulated follicular 8335/3
Site: thyroid, NOS C73.9
pw 11/22/11

Criteria	Yes	No
Diagnosis Discrepancy		
Primary Tumor Site Discrepancy	X	
HIPAA Discrepancy		
Prior Malignancy History		
Dual/Synchronous Primary Notes		
Case is (circle):	NOTIFIED	DISQUALIFIED
Reviewer Initials	Reviewed: 5/6/11	

ADDENDA:
Addendum

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Block J:

- A. Mutations in **BRAF, NRAS61, HRAS61, KRAS12/13** NOT identified.
- B. FISH test results are negative for **RET/PTC** rearrangement.

NOTE:

DNA was extracted in the amount sufficient for testing.

BACKGROUND:

Mutations in either **BRAF** or **RAS** genes or **RET/PTC** rearrangements are found in more than 70% of papillary thyroid carcinomas (1). **BRAF V600E (T1799A)** mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between **BRAF V600E** mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the **RAS** genes or **PAX8/PPARg** rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, **BRAF V600E** mutation and **RET/PTC** and **PAX8/PPARg** rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). **RAS** mutations occur in malignant and benign thyroid tumors, being found in ~40-50% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

1. Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. *Am J Surg Pathol* 2006, 30:216-222
2. Xing M, et al. BRAF mutation predicts a poorer clinical prognosis for papillary thyroid cancer. *J Clin Endocrinol Metab* 2005, 90:6373-6379.
3. Elisei R, et al. BRAF V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. *J Clin Endocrinol Metab* 2008, 93:3943-3949.
4. Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential BRAF(V600E) mutational status in high aggressive papillary thyroid microcarcinoma. *Ann Surg Oncol*. 2009 Feb;16(2):240-5.
5. Nikiforova MN, et al. RAS point mutations and PAX8-PPARγ rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. *J Clin Endocrinol Metab* 2003, 88: 2318-26.
6. Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV., ed. *Endocrine Pathology: Differential Diagnosis and Molecular Advances*. Humana Press, Totowa, 2004, 191-209.

MUTATIONAL ANALYSIS:

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the platform to amplify **BRAF** codons 599-601, **NRAS** codon 61, **HRAS** codon 61, and **KRAS** codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. If required, the mutation type was confirmed by Sanger sequencing of the PCR product on DNA from samples positive for each of these mutations was used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10⁻⁵ % of alleles with mutation present in the background of normal DNA.

FINAL DIAGNOSIS:

THYROID, TOTAL THYROIDECTOMY, 28 GRAMS -

- A. **ENCAPSULATED FOLLICULAR CARCINOMA WITH FOCI OF TRANSCAPSULAR INVASION, 2.5 CM, RIGHT LOBE.**
(See comment)
- B. **PAPILLARY MICROCARCINOMA, 0.2 CM, RIGHT LOBE.**
- C. **PAPILLARY MICROCARCINOMA, 0.8 CM, LEFT LOBE.**
- D. **CHRONIC LYMPHOCYTIC THYROIDITIS.**
- E. **ONE (1) NORMOCYLLULAR PARATHYROID GLAND ATTACHED TO THE RIGHT LOBE.**

COMMENT:

No angiolymphatic or extrathyroidal invasion is seen. Margins are free. Molecular studies will be done and reported in an addendum.

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE:	Total Thyroidectomy
TUMOR SITE:	Right Lobe ✓
TUMOR FOCALITY:	Unifocal ✓
TUMOR SIZE (largest nodule):	Greatest Dimension: 2.5 cm
HISTOLOGIC TYPE**:	Follicular Carcinoma ✓
PRIMARY TUMOR (pT):	pT2
REGIONAL LYMPH NODES (pN):	pNX
DISTANT METASTASIS (pM):	Not applicable
EXTRATHYROIDAL EXTENSION:	Not identified
MARGINS:	Margins uninvolved by carcinoma
LYMPH-VASCULAR INVASION:	Not identified
ADDITIONAL PATHOLOGIC FINDINGS:	Thyroiditis Other: Papillary microcarcinoma, 2 foci

SPECIAL PROCEDURES:

In Situ Procedure

Interpretation

Probe: *RET/PTC* Probe (10q11.2) and *RET/PTC1* Dual-Color Fusion Probe 10q11.2-10q21

Probe Description: The *RET* probe (labeled) is a 207 kb DNA probe located at band 10q11.2. The H4 and 368 probes are approximately 100kb each (labeled in), and are located at band 10q21.

Probe:
Probe Description: The *RET* probe (labeled) is a 207 kb DNA probe located at band 10q11.2. The *ELE1(RFG)* probe is approximately 100kb (labeled), and is located at band 10q11.2.

RET/PTC: The targeted area of the tissue showed 12(10.0%) of the cells with the rearrangement pattern and 108(90.0%) of the cells with the normal pattern. The rearrangement pattern of *RET* hybridization in the target area is most likely due to trisomy of chromosome 10, which was demonstrated by hybridization with CEP10, and therefore the target area is considered NOT rearranged for the *RET/PTC* region.

The *RET/PTC1* Probe did not contain the rearrangement pattern.

The *RET/PTC3* Probe did not contain the rearrangement pattern.

Results

Methodology Description:

RET/PTC rearrangement is suggested to be present when three *RET* signals are identified in greater than 8% of the analyzed cells.

RET/PTC1 rearrangement detection is as follows: Normal cells without the *RET/PTC1* rearrangement show two pair of green and orange signals separated. Cells with the *RET/PTC1* rearrangement show two pair of signals (green and orange) in juxtaposition and one green and one orange signal separated in greater than 8% of analyzed cells within the targeted area.

RET/PTC3 rearrangement detection is as follows: Normal cells without the *RET/PTC3* rearrangement show two pair of green and orange signals juxtaposition. Cells with the *RET/PTC3* rearrangement show two pair of signals (green and orange) in juxtaposition and one green and one orange signal separated in greater than 8% of analyzed cells within the targeted area.

FISH analysis was manually performed and quantitatively assessed by analysis of a minimum of 60 cells.